The Tethyan Crucible: Paleoceanographic, Climatological, and Ecological Reconstruction of the Aptian-Albian Transition (~111 Ma)

1. Introduction: The Tethyan Corridor as a Global Engine

The transition from the Aptian to the Albian stage of the Early Cretaceous, centered approximately 111 million years ago (Ma), represents one of the most dynamic and transformative intervals in the Phanerozoic Eon. This period was not merely a static snapshot of a "greenhouse earth" but a tumultuous epoch characterized by extreme climatic volatility, profound reorganization of global ocean circulation, and the fundamental restructuring of marine ecosystems. The Tethys Sea (or Neo-Tethys) served as the geographic and thermodynamic heart of this system—a vast, circum-equatorial oceanic corridor separating the dispersing fragments of Gondwana from the Eurasian landmass.

Unlike the modern Atlantic or Pacific, the Tethys was a complex "oceanic gateway" cluttered with migrating microplates, active island arcs, and shallow carbonate platforms. It functioned as the primary conduit for the redistribution of solar energy via a powerful westward-flowing circum-global current, regulating the planet's thermal gradient. However, roughly 111 million years ago, this system underwent a critical perturbation. The geological record reveals a paradoxical "cold snap" embedded within the super-greenhouse climate, coincident with severe perturbations in the carbon cycle known as Oceanic Anoxic Event 1b (OAE 1b). These events did not occur in isolation but were intrinsically linked to the tectonic dismantling of Pangea, the explosive volcanism of the Kohistan-Ladakh arc, and the evolutionary radiation of new biological architects, from the reef-building rudist bivalves to the serpentine ancestors of the mosasaurs.

This report provides an exhaustive reconstruction of the marine and estuarine environments of the Tethyan realm during this specific interval. By synthesizing high-resolution isotopic records, paleobiological data, and geodynamic modeling, we dissect the physical and biological machinery of the Aptian-Albian world. We explore the "purple oceans" poisoned by hydrogen sulfide, the hyper-energy storm systems that battered the archipelagos, and the evolutionary nurseries within the mangrove-like *Frenelopsis* forests that fringed the shallow seas.

2. Geodynamic Architecture: The Fragmented Ocean

The geography of the Early Cretaceous was defined by the progressive disintegration of the Pangean supercontinent. By 111 Ma, the separation of South America from Africa was accelerating, opening the South Atlantic, while the Tethys Ocean formed a widening wedge between the northern Laurasian continents and the southern Gondwanan blocks.

2.1 The Microplate Mosaic of the Central Tethys

The central Tethys was not a uniform expanse of open water. It was a tectonically complex region characterized by a "mosaic" of microplates, terranes, and carbonate platforms that had rifted from the northern margin of Gondwana and were drifting northward toward Eurasia. This configuration created a unique oceanographic setting resembling the modern Southeast Asian archipelago but on a continental scale.

The Adria (Apulia) Promontory

The most significant of these microplates was the Adria (or Apulia) block, a massive promontory of the African plate. During the Aptian-Albian, Adria was largely submerged, hosting the vast **Apulia Carbonate Platform**. This platform was a shallow, tropical sea floor dominated by biogenic carbonate production, separated from the African mainland by deeper pelagic basins (such as the Ionian Basin). The sedimentary record of Adria, preserved today in the limestones of Italy and Croatia, reveals a cyclic depositional environment controlled by fluctuating sea levels and orbital forcing.[1]

The isolation of the Adria platform created a natural laboratory for evolutionary experimentation. While primarily a marine domain, ephemeral land bridges or emergent islands allowed for the colonization of terrestrial fauna. Footprint assemblages from this period indicate the presence of diverse dinosaur populations, including theropods and sauropods, which likely traversed tidal flats and shallow lagoons.[3] The tectonic independence of Adria meant that its biological communities were periodically severed from the main Gondwanan populations, fostering endemism.

The Sakarya and Pontide Terranes

To the north of the Neo-Tethys lay the Sakarya and Pontide terranes, which formed the active southern margin of Eurasia (modern-day Turkey). These blocks were separated from the main Tethyan ocean by the Intra-Pontide Ocean and were sites of active subduction. The collision and accretion histories of these terranes were critical in defining the northern boundary conditions of the Tethys, regulating the exchange of water masses between the Tethys proper and the marginal basins of the Proto-Black Sea.[4]

2.2 The Kohistan-Ladakh Intra-Oceanic Arc

A defining, yet often overlooked, feature of the Neo-Tethys at 111 Ma was the

Kohistan-Ladakh Arc (KLA). This massive intra-oceanic island arc system extended for thousands of kilometers across the Tethys, situated north of the drifting Indian plate and south of the Eurasian margin. The KLA was formed by the northward subduction of Neo-Tethyan oceanic crust, creating a volcanic chain analogous to the modern Aleutian or Mariana arcs but with significantly higher magmatic output.[6]

Geological Composition and Activity:
The arc basement consisted of ultramafic and gabbroic roots, overlain by a thick sequence of calc-alkaline volcanic rocks known as the Dras Volcanics. These included pillow lavas, volcanic breccias, and pyroclastic flows of basaltic, andesitic, to rhyolitic composition.8 The presence of evolved rhyolites and dacites indicates that the arc had matured enough to develop a thickened crust (up to 55 km), capable of fractional crystallization and the generation of explosive silicic magmas.
Implications for the Tethyan Environment:
The active KLA was a primary source of environmental forcing. Frequent explosive eruptions injected massive quantities of ash, aerosols (sulfur dioxide), and CO_2 into the atmosphere.

- **Volcanic Winters:** Short-term pulses of sulfate aerosols likely contributed to intermittent cooling episodes, potentially triggering the rapid temperature fluctuations observed in the Aptian-Albian record.
- **Nutrient Seeding:** The deposition of volcanic ash into the surrounding ocean provided iron and other micronutrients, stimulating primary productivity in the Tethyan surface waters and exacerbating anoxic events.
- **"Stealth" Eruptions:** Recent modeling of similar arc volcanoes (e.g., Veniaminof) suggests that many eruptions may have occurred without significant seismic precursors, driven by the varying rheology of the magma reservoirs.[10] This implies the Tethyan environment was subject to unpredictable, sudden paroxysms of volcanic activity that would have periodically devastated local island ecosystems.

2.3 Oceanic Gateways and Circulation Patterns

The global ocean circulation at ~111 Ma was dominated by a circum-equatorial current (CEC) that flowed westward through the Tethys. Driven by the easterly trade winds, this warm, saline current transported heat from the diverse, shallow basins of the Tethys into the embryonic Central Atlantic and eventually to the Pacific.[12]

- **The Tethyan Heat Engine:** This throughflow prevented the buildup of extreme latitudinal thermal gradients, contributing to the generally equable Mesozoic climate. However, the bathymetry of the Tethys—cluttered with the microplates described above—created complex eddies, upwelling zones, and restricted sub-basins.
- **Deep Water Formation:** Unlike the modern ocean, where deep water forms in polar regions, Cretaceous deep water formation may have occurred in the warm, saline shelves of the Tethys. As water evaporated in the extensive shallow platforms (like Adria), it became dense and sank, forming Warm Saline Bottom Water (WSBW). This mechanism, however, was prone to stagnation, a key factor in the development of oceanic anoxic

events.[13]

Table 1: Tectonic and Geographic Components of the Tethys (~111 Ma)

Component	Nature/Type	Location	Paleo-Environmental Role	Evidence Source
Adria (Apulia)	Continental Microplate	Central Tethys	Extensive shallow carbonate platform; insular dinosaur habitat.	[1]
Kohistan-Ladakh	Intra-Oceanic Arc	Neo-Tethys Ocean	Source of explosive volcanism, ash, and CO$_2$; distinct biogeographic province.	[6]
Sakarya Zone	Continental Terrane	Northern Margin (Eurasia)	Active margin; barrier to northern basins.	[4]
Pindos Ocean	Deep Oceanic Basin	Western Tethys	Site of deep-water sedimentation and radiolarite deposition.	[14]
Circum-Equatorial Current	Ocean Current	Tethys-wide	Primary heat transport mechanism; westward flow.	[12]

3. Climatology: The Paradox of the "Cold Snap"

The traditional view of the Cretaceous as a monolithic, stable "super-greenhouse" has been dismantled by high-resolution proxy data. The interval around 111 Ma reveals a climate system operating at the edge of stability, oscillating between extreme warmth and sharp, transient cooling phases.

3.1 The Aptian-Albian Thermal Maximum

The baseline climate of the Early Cretaceous was undoubtedly warm. Atmospheric CO_2 concentrations are estimated to have been between 500 and 1,000 ppmv, driven by the breakup of Pangea and the high spreading rates of mid-ocean ridges.[15]

- **Sea Surface Temperatures (SST):** Oxygen isotope ($\delta^{18}O$) analysis of well-preserved rudist bivalve shells from the Tethyan subtropics (20°–30°N) yields maximum SST estimates of 35°C to 37°C.[16] These temperatures are comparable to the hottest modern tropical pools but extended over a much wider latitudinal band.
- **The Hypercane Hypothesis:** Such extreme SSTs have profound implications for atmospheric dynamics. Thermodynamic modeling suggests that when widespread SSTs exceed 30°C–35°C, the atmosphere can support "hypercanes"—tropical cyclones of significantly greater intensity and longevity than modern hurricanes. The Tethys, with its warm, extensive, and relatively enclosed waters, would have been a genesis center for these storms.[15] These hyper-storms would have deeply mixed the upper ocean, battered the coastal *Frenelopsis* forests, and transported massive amounts of water vapor and heat to higher latitudes.

3.2 The Aptian-Albian "Cold Snap"

Directly contradicting the greenhouse baseline is robust evidence for a distinct cooling interval occurring near the Aptian-Albian boundary. This "cold snap" is recorded in multiple independent proxies:

1. **Clumped Isotopes (Δ_{47}):** Data from paleosol carbonates in the Asian interior (downwind of the Tethys) indicate Mean Annual Air Temperatures (MAAT) of ~14.9°C, significantly cooler than expected for a super-greenhouse.[18]
2. **Rudist Sclerochronology:** Intra-shell isotopic variations in rudists reveal a dramatic increase in seasonality during these cooler episodes. While warm intervals showed low seasonal variance (<12°C), the cool intervals exhibit seasonal SST ranges of up to 18°C.[16]
3. **Glacial Ephemerality:** This high seasonality and lower baseline temperature are compatible with the development of ephemeral polar ice sheets or extensive alpine glaciation in the high latitudes, challenging the "ice-free" definition of the Cretaceous.[16]

Mechanism of Cooling:
The primary driver for this cooling was likely the "silicate weathering thermostat." The extreme

warmth and hydrological intensity of the preceding Early Aptian (OAE 1a interval) accelerated the chemical weathering of continental rocks. This process consumes atmospheric CO_2 and transports alkalinity to the oceans. Over millions of years, this drawdown reduced the greenhouse forcing, tipping the climate system into a transient cool phase before volcanic outgassing (from the KLA and other sources) restored high CO_2 levels.18

3.3 The Hydrological Cycle and Weathering

The hydrologic cycle at 111 Ma was supercharged. The high surface temperatures drove intense evaporation in the Tethys, while the atmospheric circulation transported this moisture to the continental interiors.

- **Isotopic Evidence:** Records of dinosaur apatite $\delta^{18}O_p$ (which reflects the isotopic composition of drinking water) show significant variability, indicating major shifts in precipitation patterns and the $\delta^{18}O$ of meteoric water.[18]
- **Sedimentary Response:** This enhanced hydrological cycle led to massive pulses of freshwater runoff and clastic sediment flux into the Tethyan basins. This freshwater influx induced salinity stratification in the marginal seas, a critical precursor to the development of anoxia.

4. Oceanography and Biogeochemistry: The Purple Ocean

The interplay between the active hydrological cycle, tectonic restriction, and high primary productivity culminated in one of the most chemically distinct states of the Tethys Ocean: **Oceanic Anoxic Event 1b (OAE 1b)**.

4.1 Stratification and Stagnation

The geometry of the Tethyan basins was particularly susceptible to stagnation. As freshwater runoff formed a low-density "lid" on the surface, and warm, saline waters formed dense bottom layers, vertical mixing was inhibited.

- **The Oxygen Minimum Zone (OMZ):** As organic matter from surface productivity sank, its decomposition consumed the available dissolved oxygen. Without vertical circulation to replenish it, the water column became dysoxic (low oxygen) and eventually anoxic (zero oxygen).[14]
- **The Jacob, Paquier, and Urbino Horizons:** These specific black shale layers mark the pulses of OAE 1b. They are rich in organic carbon and devoid of benthic fossils, testifying to the lethality of the bottom waters.[20]

4.2 Photic Zone Euxinia (PZE)

The chemical deterioration of the Tethys did not stop at anoxia. Geochemical biomarkers provide evidence for **Photic Zone Euxinia**, a condition where hydrogen sulfide

(H_2S)-laden waters rise into the sunlit surface layer.

- **Biomarker Evidence:** The presence of isorenieratane and other specific aryl isoprenoids in Tethyan sediments confirms the proliferation of **Chlorobiaceae** (green sulfur bacteria) and **Chromatiaceae** (purple sulfur bacteria).[21] These organisms are obligate anaerobes and phototrophs; they require both light for photosynthesis and hydrogen sulfide as an electron donor.
- **The Visual Consequence:** For the observer at 111 Ma, the Tethys would not have always been blue. During peak OAE 1b intervals, blooms of these bacteria would have stained the waters of restricted lagoons and basins. The water would appear a milky, distinct **purple or turquoise-pink**, contrasting sharply with the green of the *Frenelopsis* mangroves.
- **Toxicity:** The metabolism of these bacteria, and the underlying sulfate reduction, releases hydrogen sulfide gas. This "rotten egg" gas is highly toxic to aerobic life. Periodically, large bubbles of H_2S could erupt from the depths (chemocline excursions), causing massive kills of fish, ammonites, and marine reptiles, and potentially poisoning the coastal air.[21]

4.3 Tidal Resonance in Epicontinental Seas

While the deep basins stagnated, the shallow epicontinental seaways bordering the Tethys (such as the connection to the proto-North Atlantic or the Western Interior Seaway) experienced a different hydrodynamic regime. Numerical modeling indicates that the specific bathymetry of these shallow seas could amplify tidal waves, creating **resonant systems**.[24]

- **Bed Shear Stress:** In these meso-tidal to macro-tidal environments, strong tidal currents generated high bed shear stress, scouring the seabed and preventing the deposition of muds. These high-energy environments were dominated by sand ridges and dunes, inhabited by organisms adapted to shifting substrates.
- **Sedimentary Implications:** This explains the presence of cross-bedded sandstones and oolitic shoals in the Tethyan margins, alternating with the stagnant black shales of the deeper basins.

Table 2: Oceanographic States of the Aptian-Albian Tethys

State	Characteristics	Biological Indicator	Chemical Marker	Visual Appearance
Normal Marine	Oxic water column, diverse benthos.	Rudists, Foraminifera, Ammonites.	Normal $\delta^{13}C$.	Clear/Blue water.

Dysoxic	Low oxygen bottom waters.	Low-diversity benthos, stress-tolerant taxa.	Slight TOC enrichment.	Turbid/Green (algal blooms).
Anoxic	Zero oxygen bottom waters.	No benthos (laminated shales).	High TOC, Pyrite framboids.	Dark/Black bottom water.
Euxinic (PZE)	Sulfidic water in photic zone.	Green/Purple Sulfur Bacteria.	Isorenieratane, Aryl Isoprenoids.	**Purple/Milky/ Pink** surface water.

5. The Marine Biosphere: Ecosystems in Transition

The biological response to these shifting physical conditions was a dramatic turnover in marine ecosystems. The Aptian-Albian boundary marks the decline of the "Jurassic-style" coral reefs and the ascent of the rudist bivalves, as well as the diversification of plankton that would form the basis of modern marine food webs.

5.1 The Rudist Hegemony

In the shallow, warm, and often saline carbonate platforms of the Tethys (e.g., Adria, the Taurides), scleractinian corals were largely marginalized. They were replaced by **Rudist Bivalves** (Hippuritida) as the primary carbonate producers.[26]

- **Competitive Advantage:** Rudists were uniquely adapted to the "super-greenhouse" conditions. Unlike corals, which are sensitive to heat, acidification, and turbidity, rudists were robust. Their calcitic shells were more resistant to dissolution than the aragonitic skeletons of corals.
- **Morphology and Ecology:** The dominant forms in the Aptian-Albian were the "elevator" rudists, such as the **Caprinidae** and **Polyconitidae**.[27] These animals grew upright, resembling vases or cones, forming dense, gregarious thickets. They did not form a solid, wave-resistant framework like modern coral reefs but rather "meadows" of vertical shells that trapped sediment.
- **Symbiosis:** It is widely hypothesized that rudists harbored photosynthetic symbionts (zooxanthellae), similar to modern giant clams. This would explain their rapid growth rates and dominance in oligotrophic, sunlit waters.[26]

5.2 The Planktonic Revolution and Bioluminescence

The base of the Tethyan food web was undergoing an explosion of diversity.

- **Dinoflagellates:** The Early Cretaceous witnessed a massive radiation of dinoflagellates.[28] These single-celled organisms possess the cellular machinery for **bioluminescence** (the luciferin-luciferase reaction). In the nutrient-rich waters of the Tethyan upwelling zones, dinoflagellates likely formed massive blooms.
  - *The Nocturnal Seascape:* At night, the Tethys would have been spectacularly bioluminescent. Any disturbance—a swimming plesiosaur, a breaking wave, or a storm—would trigger flashes of blue-green light. This bioluminescence served as a burglar alarm defense, illuminating predators to *their* own predators.[29]
- **Calcareous Nannoplankton:** Coccolithophores were also abundant, producing the vast quantities of calcite that give the Cretaceous ("Chalk") period its name. However, during the OAE 1b acidification pulses, "biocalcification crises" occurred, temporarily suppressing these populations and favoring organic-walled plankton.[31]

5.3 Marine Reptiles: The Apex Guild

The vertebrate fauna of the Tethys at 111 Ma contradicts the older narrative that Ichthyosaurs were extinct or in terminal decline. Instead, the ecosystem supported a diverse, multi-tiered guild of marine reptiles.

Persistence of the Ichthyosaurs

The Albian Tethys remained a stronghold for the **Ophthalmosauridae**, specifically the subfamily **Platypterygiinae** (e.g., *Platypterygius*, *Sisteronia*).

- **Adaptation:** These animals were highly specialized for open-ocean cruising. They possessed large eyes for hunting in low light (or deep scattering layers) and streamlined bodies for high-speed pursuit. Tooth wear patterns indicate varied diets, from crushing ammonites and crustaceans to ram-feeding on schooling fish.[32]
- **Refutation of Extinction:** The high diversity of Albian ichthyosaurs in the Tethys (Western Europe/Russia) contrasts with lower diversity in North America, suggesting the Tethys was a critical refuge and diversification center prior to their final extinction in the Cenomanian.[32]

The Rise of Mosasauroids: Aigialosaurs and Dolichosaurs

The shallow, complex habitats created by the rudist meadows and coastal mangroves facilitated the evolution of a new group: the **Pythonomorphs**, ancestors to the true mosasaurs.

- **Aigialosaurs and Dolichosaurs:** These small (<1-2 meters), lizard-like reptiles (e.g., *Dolichosaurus*, *Coniasaurus*) were semi-aquatic. They possessed elongated bodies and reduced limbs but lacked the fully developed paddles of later mosasaurs.
- **Habitat:** They thrived in the shallow, protected lagoonal environments of the Adria and Sakarya blocks. Their serpentine bodies allowed them to navigate the dense root systems of *Frenelopsis* mangroves and the crevices of rudist thickets, ambushing small fish and

invertebrates.[34] They represent the crucial evolutionary link between terrestrial varanoids and the marine apex predators of the Late Cretaceous.[36]

Plesiosaurs and Pliosaurs

The middle and upper trophic levels were occupied by Plesiosaurs.

- **Elasmosaurids:** Long-necked forms adapted for stealthy predation on schooling fish.
- **Pliosaurs:** Large-skulled macropredators (e.g., relatives of *Kronosaurus*) remained the top predators, capable of dismembering large ichthyosaurs and turtles. They persisted in the Tethys well into the Turonian.[37]

6. Coastal and Estuarine Geobotany: The Alien Shore

The interface between the land and the Tethys Sea was occupied by floral communities that have no direct modern analog. While the "Cretaceous Terrestrial Revolution" (the rise of flowering plants) was beginning in the hinterlands, the coastline was still dominated by Gymnosperms.

6.1 The *Cheirolepidiaceae* "Mangroves"

The ecological niche of modern mangroves (salt-tolerant coastal forests) was filled by an extinct family of conifers known as the **Cheirolepidiaceae**, with the genus ***Frenelopsis*** being the most ubiquitous along the Tethyan margins.[38]

- **Morphology:** *Frenelopsis* did not look like a pine tree. It was a succulent, halophytic conifer with segmented, photosynthetic green stems and reduced, scale-like leaves. It resembled a giant, woody succulent or horsetail more than a modern conifer.[39]
- **Landscape:** These plants formed dense, monospecific thickets in the tidal flats and salt marshes. They were adapted to the high salinity and arid/seasonal conditions of the Tethyan coast. They produced massive amounts of ***Classopollis*** pollen, which often formed yellow scums on the water surface of lagoons.[38]
- **Fire Ecology:** Charcoal records indicate that these forests were prone to wildfires, likely driven by the high atmospheric oxygen and the accumulation of dry, resinous biomass during the dry seasons of the "cold snap" intervals.[41]

6.2 The Angiosperm Encroachment

In the riparian zones and disturbed habitats behind the *Frenelopsis* belt, early Angiosperms (flowering plants) were rapidly diversifying.

- **Edible Flora:** This period saw the emergence of plants producing early fruits and nuts, offering new nutritional resources for terrestrial herbivores. Although true grasses were not yet dominant, the understory of the coastal forests was becoming increasingly complex with ferns, cycads, and early broadleaf shrubs.[42]

6.3 Island Dwarfism and the Adria/Hateg Connection

The fragmented geography of the Tethys had profound effects on terrestrial evolution. The Adria microplate and other isolated terranes (which would later form parts of Romania and the Hateg Basin) were sites of **insular dwarfism**.

- **The Process:** While the famous Hateg Island fauna (dwarf titanosaurs like *Magyarosaurus*) is of Maastrichtian age, the *process* of isolation began in the Early Cretaceous. Dinosaur populations stranded on the carbonate platforms of Adria by rising sea levels were subjected to resource limitation. Over generations, large sauropods evolved smaller body sizes to survive in the restricted area of the islands. This "Island Rule" was a persistent feature of the Tethyan archipelago throughout the Cretaceous.[3]

7. Synthesis: The Kohistan-Ladakh Ecosystem Case Study

To visualize the totality of the Tethyan environment at ~111 Ma, we can reconstruct the ecosystem of the **Kohistan-Ladakh Arc**.

- **The Setting:** Imagine a chain of volcanic islands stretching across the open Neo-Tethys, 2,000 kilometers from the nearest continent. The sky is often hazy with volcanic smog (SO_2) and ash.
- **The Island:** A towering central stratovolcano, capped with snow despite the tropical latitude (due to altitude). The slopes are composed of grey andesitic lava and rhyolitic pumice.
- **The Coast:** Black sand beaches give way to rocky headlands. In the intertidal zone, *Frenelopsis* conifers cling to the rocks, their succulent stems resisting the salt spray.
- **The Reef:** Just offshore, the seabed is covered in a dense thicket of cone-shaped *Caprinid* rudists, their lids flapping open to filter feed. There are no branching corals.
- **The Water:** The water is warm (30°C) and teeming with ammonites. At night, the breaking waves glow neon blue from dinoflagellates.
- **The Danger:** A 3-meter *Platypterygius* ichthyosaur patrols the reef edge. In the shallows, a small, snake-like *Dolichosaurus* weaves through the rudist shells, hunting shrimp.
- **The Catastrophe:** Suddenly, without seismic warning ("stealth eruption"), the volcano flanks rupture. Pyroclastic flows race down the slopes, boiling the sea surface. The ash column collapses, sending a turbidity current into the deep trench, burying the anoxic bacterial mats in the abyss.

8. Conclusion

The Tethys Sea of ~111 Ma was a realm of dynamic contradictions. It was the Earth's thermostat, yet it was prone to violent fluctuations between super-greenhouse heat and sudden cooling. It was a biological cradle, fostering the rise of the mosasaurs and the dominance of rudists, yet it was also a chemical killing field, where the waters could turn purple and poisonous with hydrogen sulfide.

This era serves as a potent analog for our own future. The Aptian-Albian demonstrates that in a high-CO$_2$ world, the ocean is not a stable buffer but a volatile system capable of rapid deoxygenation, extreme storm generation, and fundamental ecosystem shifts. The persistence of life in the Tethys—from the bioluminescent plankton to the island-hopping dinosaurs—is a testament to resilience, but the fossil record clearly shows that such resilience comes at the cost of catastrophic turnover and extinction.

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